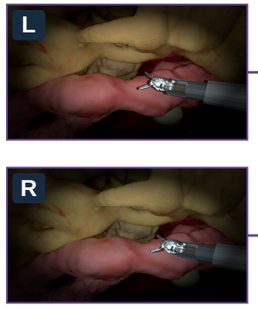
Foreground-guided stereo-temporal articulated pose estimation

Shared visual features • Three task-specific branches • Pose in each camera coordinate system

Stereo RGB pair



1280 × 720 / eye

SHARED VISUAL ENCODER

ConvNeXtV2 Nano

L/R weight sharing
640 × 360 input

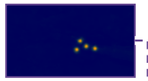
128-channel FPN

Feature map [N, 128, 90, 160]

Predicted localization



1 mask



5 keypoints

Each map: 90 × 160

L F C H1 H2 J1 J2

R F C H1 H2 J1 J2

6 × 128 tokens / eye

+ camera / view embeddings

Original-resolution RGB

Foreground ROIs

Detached mask / boxes



L R



L R

SHAFT / BODY

Fused C token (128)

previous state

Cross-view attention
2 blocks · 4 heads
[BT, 2, 6, 128]

Body GRU
[2B, T, 128]
→ [2B, T, 256]

Rotation + depth
R6D (6) + Z (1)
C from pixel ray

14

Pool: 6 × 128 → 128

Two camera-frame poses: C + quaternion

WRIST

Branch-specific weights; shared across L/R.

previous state

Local CNN
4 × 160 × 160
→ 128-D

m

Stereo MLP
[L | R], [R | L]
256 → 128

Wrist GRU
128 → 256
per-eye state

Angle head
[hL | hR]: 512
256 → 2

2

PAIRED JAW

Branch-specific weights; shared across L/R.

previous state

Local CNN
4 × 160 × 160
→ 128-D

m

Stereo MLP
[L | R], [R | L]
256 → 128

Jaw GRU
128 → 256
per-eye state

Angle head
[hL | hR]: 512
256 → 1

1

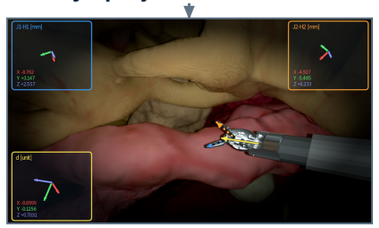
ARTICULATED POSE [B, T, 17]

Left: C (3) + quaternion (4)
Right: C (3) + quaternion (4)
Wrist pitch + yaw (2)
Jaw half-opening angle (1)

Instrument kinematics

d, H1/H2, J1/J2, T, local XYZ

Per-eye projection with K and D



Three vectors + local XYZ panels

CAMERA CONDITIONING

Known K + Brown distortion D

Global token embedding
9 → 128 dimensions

Local metadata m
Camera (9) + ROI (4)
+ mask fraction (1)
+ keypoint confidence (1)

15 → 128 dimensions

No stereo extrinsic input

C FROM A SINGLE RAY

Predicted C pixel
+ K, D → undistorted ray

$$C = Z\,[x_n, y_n, 1]$$

Learn Z

C

X, Y follow the predicted ray.

No second lateral regression.

F: foreground token H1/H2: hinges J1/J2: jaw tips

Foreground ROIs use the original 720p images.

N = 2BT • B: batch T: time • Training T = 8 • Streaming T = 1 + history • Three GRU states: [3, 2B, 256]

stereo_pose_unified_c_v1